

Residual-Shifted Diffusion Models for Efficient Label-Free Cell Painting

Reproducing and Extending CellResDM with Consistency Distillation

A REPORT PRESENTED

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Abstract

High-content phenotypic screening, and especially the Cell Painting assay, has revolutionized early-stage drug discovery by enabling rapid morphological profiling of cells under thousands of perturbations. However, fluorescence-based imaging is costly, time-consuming, and prone to artefacts, limiting its scalability. Computational prediction of immunofluorescence (IF) from label-free bright-field (BF) microscopy offers a promising alternative, but existing convolutional and adversarial models suffer from blurred outputs or unstable training, while diffusion models incur high inference costs.

Here, we reproduce and extend the CellResDM framework—a residual-shifted diffusion model that efficiently predicts five-channel Cell Painting IF from a single BF image in as few as 15 steps, over $60\times$ faster than standard diffusion, with strong fidelity to both pixel-level and single-cell morphological features. We further introduce Consistency Distilled CellResDM, integrating consistency distillation to reduce inference to just two steps, surpassing all baselines in accuracy and efficiency on the cpg0000-jump-pilot dataset. These advances move label-free, molecularly-resolved cell profiling closer to practical routine use in large-scale drug discovery.

1 Introduction

High-content, phenotypic high-throughput screening (HTS) is now foundational in early-stage drug discovery, enabling systematic, large-scale profiling of how thousands of chemical and genetic perturbations alter cell behaviour. Among these, the *Cell Painting* assay [1, 3] has emerged as a gold standard for unbiased morphological profiling. By multiplexed staining of key subcellular structures—nucleus, endoplasmic reticulum (ER), actin/Golgi/plasma membrane (AGP), mitochondria, and RNA—Cell Painting captures and quantifies the morphological and phenotypic features of cells. This provides rich information, making them invaluable for single-cell profiling, compound clustering, and hit triage in both academic and industrial screens.

Despite its broad utility, fluorescence imaging presents **significant practical limitations**: (i) *Cost and availability*—specialised dyes (e.g., TOTO-3 for DNA) can cost over \$500 per 384-well plate and are prone to shortages; (ii) *Throughput*—capturing five high-resolution channels per plate adds hours to each experiment and is labour-intensive; and (iii) *Phototoxicity and artefacts*—fluorescent labels may perturb cell physiology and introduce unwanted variability, complicating downstream analyses.

These limitations have spurred interest in **label-free alternatives**, foremost among them *bright-field (BF) microscopy*. BF imaging rapidly captures structural information from unstained, live cells

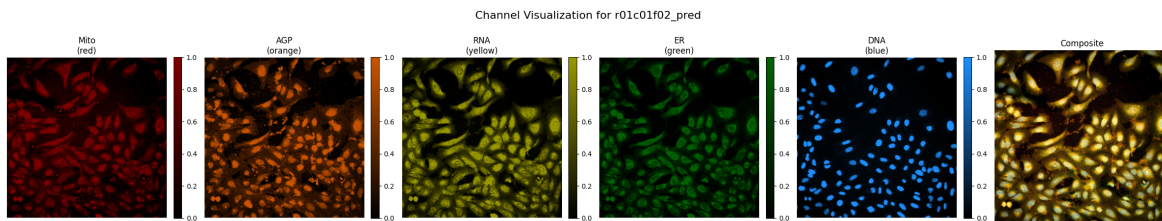


Figure 1: **Five-channel Cell Painting field-of-view.** Each panel shows a single fluorescence channel (mitochondria, AGP, RNA, ER, DNA), with the rightmost panel merging all channels into a pseudo-colour composite constructed for visualization reused throughout [3].

using transmitted light, providing inexpensive, dye-free, and non-destructive snapshots across entire screening plates. However, BF inherently lacks the molecular specificity of fluorescence: fine protein localization and many functional compartments are invisible, limiting its use for deep mechanistic profiling.

With the recent advancement in computer vision, the possibility of computationally inferring fluorescence-like molecular information from bright-field images has become increasingly feasible—and is now an area of active and growing research. Recent machine learning work has begun to address this gap: early convolutional neural networks (CNNs) [4] achieved promising reconstructions of nuclear and cytoskeletal markers; GANs [11, 12, 20] improved image sharpness; and state-of-the-art diffusion models [5, 6, 9, 23] have further improved realism. Despite these advances, each of these approaches faces notable limitations: CNN-based methods tend to oversmooth or blur fine details; GANs, while generating sharper images, are prone to instability and may hallucinate biologically implausible features; and diffusion models, despite their impressive realism, are computationally expensive and can be difficult to train at scale. Additionally, many prior studies have not thoroughly assessed whether the generated images preserve *single-cell morphological statistics*—the very features essential for robust screening, mechanistic interpretation, and downstream analysis and usage. To address these gaps, [23] proposed **CellResDM**, an efficient residual-shifted diffusion model for five-channel Cell Painting synthesis from bright-field images. CellResDM stands out for its ability to balance accuracy and computational efficiency, delivering state-of-the-art performance on the challenging Cell Painting benchmark, and was further showcased and evaluated on the Morphological Profiling task, demonstrating strong correlation with real single-cell features.

Therefore, in this work, we rigorously reproduce and validate CellResDM, and further extend it with **Consistency Distilled CellResDM**, which applies consistency distillation to accelerate inference and enhance fidelity. On the cpg0000-jump-pilot dataset, our models achieve state-of-the-art pixel accuracy and single-cell feature correlation, reducing inference steps from 15 to just 2 while outperforming all tested baselines. Together, these advances bring label-free, molecular-level cell profiling closer to practical, scalable use in drug discovery—cutting costs, eliminating dye artefacts, and crucially, preserving the biological information needed for downstream analytics.

2 Methods

2.1 Overview

We focus on the CellResDM framework, an efficient residual-shifted diffusion model for translating label-free bright-field images into multi-channel immunofluorescence Cell Painting images. Standard diffusion models generate images by de-noising from pure noise, essentially constructing an output “from scratch.” While this approach—requiring many inference steps—is powerful for creative tasks that benefit from generating entirely novel and diverse images, it is unnecessarily complex when there is already a strong structural correspondence between the input and target. To exploit the inherent structural similarity between BF and IF images, CellResDM is designed to predict only subtle adjustments—removing minor noise and correcting the residual differences between modalities. Rather than constructing the output from scratch, the model focuses on refining the BF input, learning the minimal transformations required to approximate its corresponding IF image. This targeted approach dramatically reduces the task complexity, allowing for accurate, high-quality synthesis with as few as 15 inference steps.

Building on this foundation, we incorporate consistency distillation—yielding Consistency Distilled CellResDM—which further compresses the denoising trajectory to just two inference steps, with no

sacrifice in accuracy or feature preservation. We rigorously compare both models against baseline methods and provide comprehensive evaluation using both image-level and biologically meaningful single-cell metrics.

The remainder of this section details the mathematical formulation of the diffusion process for CellResDM, the noise scheduling strategy, the integration of consistency distillation, and the experimental settings used for this work.

2.2 CellResDM: Residual-Shifted Diffusion Model

Given a target immunofluorescence (IF) image x_0 and a source bright-field (BF) image y_0 , we define the residual:

$$e_0 = y_0 - x_0$$

which encodes the pixel-wise difference between BF and IF images.

CellResDM uses this residual and diffusion framework to bridge bright-field and immunofluorescence images. For the diffusion’s forward process, it gradually adds noise and a fraction of the residual difference to the target IF image, progressively blending it with the BF modality. The reverse process then learns to iteratively remove this noise and residual, ultimately recovering the IF image conditioned on the BF input. The full process is illustrated in Figure 2, which shows how noise and residuals are incorporated at each diffusion step.

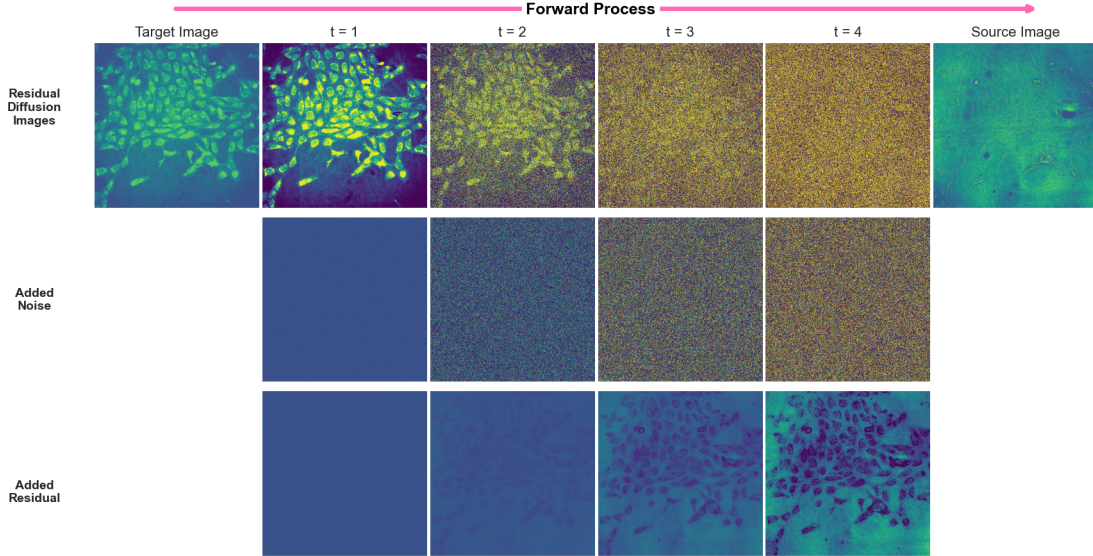


Figure 2: Visualization of the residual diffusion process between target (mitochondrial) and source (BF) image over a 4-step diffusion. Top: residual diffusion at each step. Middle: noise added at each step. Bottom: residuals.

Forward Process. Let $\{\eta_t\}_{t=0}^T$ be a monotonically increasing sequence of coefficients, with $\eta_0 = 0$, $\eta_T \approx 1$. The forward process, at each step, shifts the current image by a fraction of the residual and adds noise:

$$q(x_t|x_{t-1}, y_0) = \mathcal{N}(x_t; x_{t-1} + \alpha_t e_0, \kappa^2 \alpha_t I) \quad (1)$$

where $\alpha_t = \eta_t - \eta_{t-1}$, and κ sets the relative noise scale.

This Markovian definition fully explains our forward process, but it's inefficient for directly sampling x_t during training. By induction (see Appendix A.1), the process can be re-parametrized as:

$$q(x_t | x_0, y_0) = \mathcal{N}(x_t; x_0 + \eta_t e_0, \kappa^2 \eta_t I) \quad (2)$$

That is, for any t , we can write:

$$x_t = x_0 + \eta_t e_0 + \kappa \sqrt{\eta_t} \epsilon$$

with $\epsilon \sim \mathcal{N}(0, I)$. By expressing x_t in this way, we can bypass the sequential forward process and directly generate noisy samples at arbitrary steps, making training much more efficient.

Reverse Process. Given a fully noised sample x_T , our goal for the reverse process is to recover the target image x_0 conditioned on the clean source y_0 . This is achieved by learning a reverse Markov chain, which iteratively de-noises x_t to x_{t-1} , ultimately reconstructing x_0 . The marginal reverse generative process can be written as:

$$p(x_0 | y_0) = \int p(x_T | y_0) \prod_{t=1}^T p_\theta(x_{t-1} | x_t, y_0) dx_{1:T} \quad (3)$$

where the prior at the final time-step is a fully noised Gaussian centred on the source image:

$$p(x_T | y_0) \approx \mathcal{N}(x_T; y_0, \kappa^2 I).$$

Unlike the forward process, these reverse transitions do not follow directly from the diffusion construction and must be learned. Therefore, the reverse transitions themselves can be modelled as learned Gaussians:

$$p_\theta(x_{t-1} | x_t, y_0) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, y_0, t), \Sigma_\theta(x_t, y_0, t)), \quad (4)$$

where μ_θ and Σ_θ are predicted by a neural network.

By leveraging Bayes' theorem and the Markov property of the forward process (see Appendix A.2), the reverse posterior can be computed in closed form as:

$$q(x_{t-1} | x_t, x_0, y_0) = \mathcal{N}(x_{t-1}; \mu_t, \Sigma_t) \quad (5)$$

$$\mu_t = \frac{\eta_{t-1}}{\eta_t} x_t + \frac{\alpha_t}{\eta_t} x_0 \quad \Sigma_t = \kappa^2 \frac{\eta_{t-1}}{\eta_t} \alpha_t I \quad (6)$$

Under this construction, only x_0 is unknown during inference, so we replace x_0 in the mean with a neural network prediction $f_\theta(x_t, y_0, t)$:

$$p_\theta(x_{t-1} | x_t, y_0) = \mathcal{N}\left(x_{t-1}; \frac{\eta_{t-1}}{\eta_t} x_t + \frac{\alpha_t}{\eta_t} f_\theta(x_t, y_0, t), \kappa^2 \frac{\eta_{t-1}}{\eta_t} \alpha_t I\right) \quad (7)$$

Therefore, in CellResDM, the neural network is trained to directly predict x_0 , the target immunofluorescence image, conditioned on the current noised image x_t , the source BF image y_0 , and the timestep t . At each step of the reverse process, this prediction is blended with the current residual-noised image x_t using the weights defined in Equation 6, yielding a progressively denoised, residual-shifted image. Through this iterative procedure, the model refines x_t at every step, ultimately reconstructing a clean estimate of the true x_0 .

Training Loss. The model is trained by minimizing the evidence lower bound (ELBO), which decomposes into a sum of KL divergences between the true and learned reverse transitions at each step:

$$\min_{\theta} \sum_{t=1}^T D_{\text{KL}}(q(x_{t-1} \mid x_t, x_0, y_0) \parallel p_{\theta}(x_{t-1} \mid x_t, y_0))$$

Because both conditionals are Gaussians with the same covariance, the KL divergence reduces to a weighted mean squared error (see Appendix A.3 for the full derivation):

$$D_{\text{KL}}(q \parallel p_{\theta}) = \frac{\alpha_t}{2\kappa^2\eta_{t-1}\eta_t} \|x_0 - f_{\theta}(x_t, y_0, t)\|_2^2 \quad (8)$$

However, the coefficient $\frac{\alpha_t}{2\kappa^2\eta_{t-1}\eta_t}$ becomes extremely large for small t , leading to training instability. Therefore, following prior work, we omit the coefficient and optimize an unweighted MSE loss:

$$\mathcal{L}_{\text{CellResDM}}(\theta) = \mathbb{E}_{x_0, y_0, \epsilon, t} [\|x_0 - f_{\theta}(x_t, y_0, t)\|_2^2] \quad (9)$$

where $x_t = x_0 + \eta_t e_0 + \kappa\sqrt{\eta_t}\epsilon$. For enhanced quality, we combine MSE with the learned perceptual image patch similarity (LPIPS) loss [?]. LPIPS measures perceptual similarity by comparing the deep feature activations of two images, making it sensitive to structural and textural differences that are perceptually meaningful. This encourages the network to produce outputs that are not only pixel-wise accurate but also structurally realistic. Formally, for two images x and $\hat{x}$,

$$\mathcal{L}_{\text{LPIPS}}(x, \hat{x}) = \sum_l w_l \|\phi_l(x) - \phi_l(\hat{x})\|_2^2, \quad (10)$$

where ϕ_l denotes the feature map from layer l of a pretrained network, and w_l are learned weights reflecting perceptual importance. Therefore, our final loss can be described as:

$$\mathcal{L}_{\text{CellResDM}}(\theta) = \mathbb{E}_{x_0, y_0, \epsilon, t} [\|x_0 - f_{\theta}(x_t, y_0, t)\|_2^2 + \mathcal{L}_{\text{LPIPS}}(x_0, f_{\theta}(x_t, y_0, t))] . \quad (11)$$

2.3 Noise Scheduling

Having introduced both the forward and reverse processes, we now detail the key hyperparameters that control the addition of noise and residual in CellResDM: the sequence of noise coefficients $\{\eta_t\}_{t=0}^T$. These coefficients are fundamental, as they determine both the magnitude and schedule of residual injection and stochastic perturbation at each step of the diffusion process (see Eq. 2).

Recall that, for any t , the forward process constructs noised samples as:

$$x_t = x_0 + \eta_t e_0 + \kappa\sqrt{\eta_t}\epsilon$$

where $e_0 = y_0 - x_0$ is the fixed residual, $\epsilon \sim \mathcal{N}(0, I)$, and κ is a global noise scale. The sequence $\{\eta_t\}$ smoothly interpolates from “no residual, no noise” ($t = 0$) to “full residual, maximal noise” ($t = T$).

Boundary Conditions and Design Principles. For the scheduling to be effective and numerically stable, it must satisfy several desiderata:

- **Start near the target:** At $t = 1$, $\eta_1 \rightarrow 0$, so $x_1 \approx x_0$ + minuscule noise, ensuring the first denoising step truly reflects the data distribution’s score [18].

- **End near the source:** At $t = T$, $\eta_T \rightarrow 1$, so $x_T \approx y_0 + \text{maximal noise}$, meaning the initial state for reverse diffusion is nearly pure noise centered on the BF image.
- **Avoid degeneracies:** η_1 and η_T cannot be exactly 0 or 1. If $\eta_1 = 0$, no noise is injected, resulting in a Dirac delta and loss of generalization. If $\eta_T = 1$, the last forward step collapses, destabilizing training and inference. Additionally, $\alpha_T = \eta_T - \eta_{T-1}$ becomes very small, “killing” the model update in the last reverse step.

Therefore, for practical and numerical reasons, we adopt:

$$\eta_1 = \min \left(\left(\frac{0.04}{\kappa} \right)^2, 0.001 \right), \quad \eta_T = 0.99$$

Both are chosen to be very close to the theoretical limits, but always non-degenerate.

Geometric Noise Scheduling. Defining just the endpoints is insufficient; the full sequence $\{\eta_t\}_{t=1}^T$ crucially affects learning. Following [23], we use a non-uniform geometric progression for the *square roots* of the noise coefficients:

$$\sqrt{\eta_t} = \sqrt{\eta_1} \cdot b_0^{\beta_t}, \quad t = 2, \dots, T-1 \quad (12)$$

$$\beta_t = \left(\frac{t-1}{T-1} \right)^p \cdot (T-1), \quad b_0 = \exp \left[\frac{1}{2(T-1)} \log \frac{\eta_T}{\eta_1} \right] \quad (13)$$

where p is a shape parameter (default $p = 0.3$), controlling how aggressively the schedule increases. This schedule starts with almost no noise for early steps, ramping up steeply towards the end.

Numerical Example. For our main experiments, we used $T = 15$, $\kappa = 2.0$, $p = 0.3$, and $\eta_T = 0.99$. These values were chosen for stability and best empirical trade-off between fidelity and diversity.

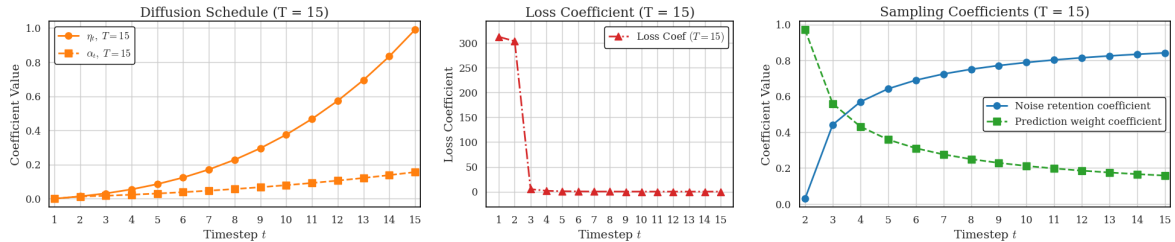


Figure 3: **Noise scheduling in CellResDM.** (Left) Training-time values of η_t and α_t , showing how noise and residual is added non-linearly over time. (Middle) Per-step loss coefficients during training, illustrating the high initial values that may cause training instability. (Right) Inference-time sampling coefficients: noise retention coefficient (η_{t-1}/η_t) and prediction coefficient (α_t/η_t), which modulate the exploration-exploitation balance at each denoising step.

The non-uniform geometric schedule for η_t is central to both inference and training dynamics in CellResDM. During inference, two coefficients—noise retention coefficient (η_{t-1}/η_t) and prediction coefficient (α_t/η_t)—control the balance between preserving prior noised residual x_t and integrating the network’s prediction of x_0 at each step. Early in the process, η_{t-1}/η_t is high, allowing for broad exploration of possible solutions due to minimal correction from the network’s prediction. As inference progresses to smaller time-steps, the prediction coefficient increases, giving greater weight to the

network’s output and enabling the model to focus on refining details and removing noise. This dynamic facilitates a smooth transition from exploratory sampling to high-fidelity reconstruction as the de-noising process unfolds.

During training, the same schedule dictates the amount of residual and noise added at each step, from subtle perturbations at early timesteps (requiring fine correction) to heavily corrupted inputs at later steps (demanding robust coarse reconstruction). One drawback of this schedule is that the associated loss coefficient, $\frac{\alpha_t}{2\kappa^2\eta_{t-1}\eta_t}$, becomes extremely large at early, low-noise steps, heavily penalizing minor mistakes and potentially destabilizing training. To address this, we follow prior work and adopt a uniform loss weighting across all timesteps during optimization.

3 Accelerating Inference via Consistency Distillation

While residual diffusion reduces the number of de-noising steps from around 1,000 in standard DDPMs [9] to just 15, inference remains relatively slow—for example, generating 13,800 images still takes about eight hours on an A100 GPU. To further accelerate inference and make residual diffusion competitive with rapid, non-diffusion approaches, we turn to *consistency models* [17].

Consistency models provide a principled framework for enabling diffusion models to generate high-quality outputs in significantly fewer steps—potentially even a single step—by encouraging the model’s predictions to be invariant across all de-noising stages. Crucially, residual diffusion fits neatly within this framework, satisfying all theoretical requirements for consistency training without requiring architectural changes (see Section 3.1).

Our motivation for adopting consistency distillation is twofold. First, it enables dramatic acceleration of the inference process, reducing computational demands and runtime. Second, we observe that predictions at different time-steps in residual diffusion can capture complementary aspects of the target image: intermediate steps may better retain local structures, while later steps can suppress artifacts but sometimes over-smooth fine details (see Figure 4). By enforcing the outputs at different time-steps to be the same, consistency distillation can combine these strengths, potentially yielding images that are both higher quality and more robust. Therefore, we expect consistency distillation to deliver improvements in both efficiency and output fidelity for residual diffusion models.

Building on these insights, we now turn to a formal definition of consistency models and show how they integrate seamlessly with residual diffusion.

3.1 Formal Definition of Consistency Models

A core property of consistency models is **self-consistency**: the model’s prediction should remain unchanged, regardless of the diffusion time-step along a given trajectory. In other words, at any point in the reverse process, the de-noised output for the clean image should be identical. This property allows the de-noising path to be dramatically shortened—even to a single step—since all intermediate predictions are aligned and thus redundant.

Figure 5 illustrates this concept. The model is trained so that its predictions at different time-steps (represented as arrows converging to a single point) always have the same result, no matter the specific path taken through the diffusion process.

Formally, following [17], let $\{\mathbf{x}_t\}_{t \in [\epsilon, T]}$ denote the solution trajectory. A function

$$f : (\mathbf{x}_t, t) \mapsto \mathbf{x}_\epsilon$$

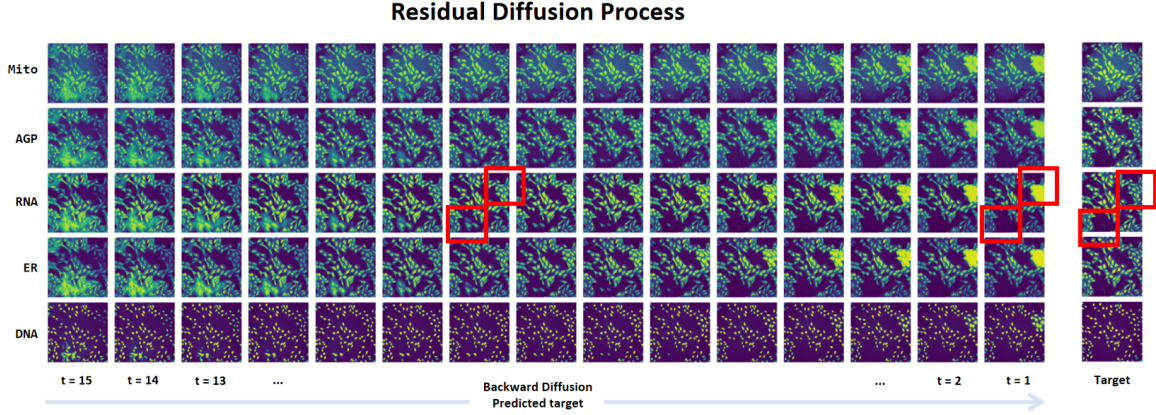


Figure 4: **Residual diffusion denoising over 15 reverse steps.** Columns show the model’s prediction of the clean image x_0 at each reverse-diffusion timestep ($t = 15$ on the left $\rightarrow t = 1$ on the right), with the ground-truth target in the final column. Rows correspond to the five Cell Painting channels (Mito, AGP, RNA, ER, DNA). In the RNA channel, red boxes highlight the difference between step 7 and step 1 compared to the target: step 7 reconstructs the upper-right region but introduces an artifact in the lower-left, while step 1 removes the artifact but slightly over-denoises the upper-right. This complementary error pattern motivates *consistency distillation*, which encourages intermediate and final predictions to agree and enables a shorter reverse trajectory.

is **self-consistent** if

$$f(\mathbf{x}_t, t) = f(\mathbf{x}_{t'}, t'), \quad \forall t, t' \in [\epsilon, T].$$

This behaviour is essential for efficient, step-agnostic inference.

To guarantee self-consistency, the model must also satisfy a *boundary condition*: it should act as the identity on nearly clean inputs,

$$f(\mathbf{x}_\epsilon, \epsilon) = \mathbf{x}_\epsilon,$$

so that already de-noised samples remain unchanged.

There are two common parametrizations that ensure this boundary condition:

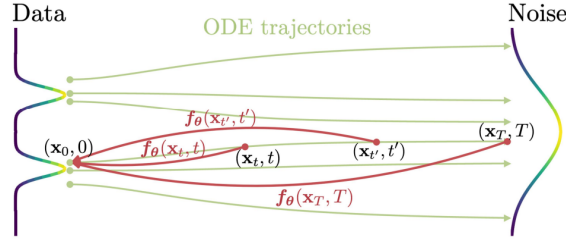


Figure 2: **Consistency models** are trained to map points on any trajectory of the **PF ODE** to the trajectory’s origin.

Figure 5: Illustration of the self-consistency property in consistency models. Regardless of the denoising trajectory or time-step, the model’s output converges to the same prediction, enabling step-agnostic inference.

- **Explicit Identity at Initial Time-step:**

$$f_{\theta}(\mathbf{x}, t) = \begin{cases} \mathbf{x}, & t = \epsilon \\ F_{\theta}(\mathbf{x}, t), & t > \epsilon \end{cases}$$

- **Skip-Connection Parametrization:**

$$f_{\theta}(\mathbf{x}, t) = c_{\text{skip}}(t) \mathbf{x} + c_{\text{out}}(t) F_{\theta}(\mathbf{x}, t),$$

where $c_{\text{skip}}(\epsilon) = 1$ and $c_{\text{out}}(\epsilon) = 0$.

In our residual diffusion model, the reverse process naturally follows the skip-connection parametrization and satisfies the boundary condition, aside from a negligible noise term. Specifically, at the initial step ($t = 1$), we have $\frac{\eta_{t-1}}{\eta_t} = 1$ and $\frac{\alpha_t}{\eta_t} = 0$, ensuring the model output exactly matches its input—thus realizing the identity mapping required for consistency models. Although our formulation includes a small noise term, our goal in cell image generation is deterministic translation between modalities. Therefore, when training consistency models, we omit the noise term in Equation 6, resulting in stable, reproducible outputs and full compatibility with the consistency framework.

Thus, the structure of our residual diffusion model—including its reverse process and noise scheduling—naturally meets both the theoretical and practical requirements for consistency modelling.

Remark on Noise Scheduling and Minimal Steps. It is important to note that, in principle, the noise schedule in consistency models may be chosen arbitrarily, and our particular schedule is fully compatible with the consistency framework. However, the specific choice of $\eta_1 \rightarrow 0$ (an almost noiseless starting point) and $\eta_T \rightarrow 1$ (a maximally noisy endpoint) introduces a practical constraint: it precludes true single-step denoising. The boundary condition at $\eta_1 \rightarrow 0$ yields a nearly clean starting sample, while $\eta_T \rightarrow 1$ corresponds to the most degraded state. As a result, at least two de-noising steps are required to traverse the entire trajectory from source to target.

While this restriction does not strictly permit single-step prediction, it nonetheless enables a reduction in the number of de-noising steps from 15 to only 2, representing a substantial acceleration over the original diffusion process.

On Adapting Multistep Sampling for Cell Images. For multi-step inference with consistency models, a greedy search to optimize the Fréchet Inception Distance (FID) metric across multi-step predictions is employed in the original paper [8, 17]. However, in the context of cell image generation, the preservation of morphological features and structural integrity is paramount—qualities that are not adequately captured by FID. Therefore, we adhere to the original residual diffusion reverse process rather than optimize for FID, as this approach is more consistent with the scientific objectives and qualitative priorities of cell imaging.

3.2 Consistency Distillation Loss

While residual diffusion is naturally compatible with self-consistency, this property must be explicitly enforced during training. To achieve this, we use the consistency distillation loss, which distills a pre-trained model’s knowledge and encourages its predictions to remain consistent across different time-steps. We adopt the following objective from [17]:

Definition 1 (Consistency Distillation Loss). *The consistency distillation loss is defined as*

$$\mathcal{L}_{\text{CD}}(\theta, \theta^-; \varphi) := \mathbb{E} \left[\lambda(t_n) d(f_\theta(x_{t_{n+1}}, t_{n+1}), f_{\theta^-}(\hat{x}_\varphi^{t_n}, t_n)) \right],$$

where the expectation is taken over $x \sim p_{\text{data}}$, $n \sim U\{1, T-1\}$, and $x_{t_{n+1}} \sim \mathcal{N}(x; t_{n+1}I)$. Here, $U\{1, T-1\}$ denotes the uniform distribution over $\{1, 2, \dots, T-1\}$; $\lambda(\cdot) \in \mathbb{R}_+$ is a positive weighting function which we here choose to be 1 for simplicity; and $d(\cdot, \cdot)$ is a non-negative metric, with $d(x, y) = 0$ if and only if $x = y$. The variable $\hat{x}_\varphi^{t_n}$ is defined as in Equation 14; θ^- denotes the exponential moving average (EMA) of past θ values, and f_θ is the current consistency model.

Definition 2 (Consistency Distillation Loss). *The consistency distillation loss is defined as*

$$\mathcal{L}_{\text{CD}}(\theta, \theta^-; \varphi) := \mathbb{E} \left[\lambda(t_n) d(f_\theta(x_{t_{n+1}}, t_{n+1}), f_{\theta^-}(\hat{x}_\varphi^{t_n}, t_n)) \right],$$

where the expectation is taken over $x \sim p_{\text{data}}$, $n \sim U\{1, T-1\}$, and $x_{t_{n+1}} \sim \mathcal{N}(x; t_{n+1}I)$. Here, $U\{1, T-1\}$ denotes the uniform distribution over $\{1, 2, \dots, T-1\}$; $\lambda(\cdot) \in \mathbb{R}_+$ is a positive weighting function (set to 1 for simplicity in our case); and $d(\cdot, \cdot)$ is a non-negative distance metric, with $d(x, y) = 0$ if and only if $x = y$. The variable $\hat{x}_\varphi^{t_n}$ is defined in Equation 14. Here, θ^- denotes the exponential moving average (EMA) of previous θ values, and f_θ is the current consistency model.

Algorithm 1 Consistency Distillation for Residual Diffusion

Require: Dataset $\mathcal{D}$, model parameters θ , learning rate η , noise schedule $\{\eta_t, \alpha_t\}$, distance metric $d(\cdot, \cdot)$, weighting function $\lambda(\cdot)$, EMA decay rate μ

- 1: Initialize EMA teacher: $\theta^- \leftarrow \theta$
 - 2: **while** not converged **do**
 - 3: Sample $x \sim \mathcal{D}$
 - 4: Sample $n \sim \mathcal{U}\{1, \dots, T-1\}$
 - 5: Sample $x_{t_{n+1}} \sim \mathcal{N}(x; \eta_{t_{n+1}}^2 I)$
 - 6: Compute $\hat{x}_\varphi^{t_n} \leftarrow \frac{\eta_{t_n}}{\eta_{t_{n+1}}} x_{t_{n+1}} + \frac{\alpha_{t_{n+1}}}{\eta_{t_{n+1}}} f_\varphi(x_{t_{n+1}}, t_{n+1})$
 - 7: Compute loss: $\mathcal{L} \leftarrow \mathcal{L}_{\text{CD}}(f_\theta(x_{t_{n+1}}, t_{n+1}), f_{\theta^-}(\hat{x}_\varphi^{t_n}, t_n))$
 - 8: Update model parameters: $\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}$
 - 9: Update EMA teacher: $\theta^- \leftarrow \text{stopgrad}(\mu \theta^- + (1 - \mu) \theta)$
 - 10: **end while**
-

Intuitively, this loss encourages the current model f_θ to match the predictions of its EMA counterpart θ^- at two adjacent time steps. When trained in batches across different timesteps, this objective enforces self-consistency throughout the denoising trajectory. The use of the EMA model helps prevent mode collapse, especially when ground-truth cell images are not available for Consistency Distillation training [7, 19].

For the derivation of $\hat{x}_\varphi^{t_n}$, the original paper proposes

$$\hat{x}_\varphi^{t_n} := x_{t_{n+1}} + (t_n - t_{n+1}) \Phi(x_{t_{n+1}}, t_{n+1}; \varphi), \quad (14)$$

where Φ denotes the teacher model and serves as the update function for a one-step ODE solver applied to the empirical PF ODE. Since we employ a different noise schedule than what is proposed in consistency model, we adapt this update using the rule in Equation 6 to recover the previous step.

Therefore, training for consistency distillation involves three models: f_θ , the EMA model f_{θ^-} , and an original reference model to generate $\hat{x}_\varphi^{t_n}$ for each batch. All models are initialized with the CellResDM pre-trained checkpoint. The entire training pipeline is outlined in Algorithm 1.

4 Evaluation Metrics

To rigorously evaluate the quality of BF-to-IF results, we devise a comprehensive suite of metrics that assess both standard image fidelity and biological validity. While conventional image metrics—such as Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM)—quantify pixel-wise and perceptual similarity, they do not directly reflect preservation of biologically meaningful features. Therefore, we supplement these with domain-specific evaluation derived from single-cell morphological features, extracted via CellProfiler¹, to ensure the synthetic images are useful for downstream scientific applications.

4.1 Standard Image Quality Metrics

We begin with widely adopted measures of pixel-level and perceptual similarity:

MSE:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (X_i - \hat{X}_i)^2$$

where N is the number of pixels, X_i is the intensity of the i -th pixel in the ground truth, and $\hat{X}_i$ is the corresponding prediction. Lower MSE values indicate better average fidelity.

PSNR:

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right)$$

where MAX_I is the maximum possible pixel value (e.g., 255 for 8-bit images). Higher PSNR reflects lower noise and better reconstruction.

SSIM:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

with μ_x , μ_y the local means, σ_x^2 , σ_y^2 the variances, σ_{xy} the local covariance, and C_1 , C_2 constants for stability. SSIM, ranging from 0 to 1, captures perceptual structure including edges and textures, making it particularly sensitive to cell boundaries.

Limitations of Fréchet Inception Distance (FID) for Cell Images Despite FID’s popularity in generative modelling for natural images, it is fundamentally ill-suited for cell microscopy applications. FID rely on feature representations learned by Inception networks trained on natural image datasets, which do not reflect the structure, texture, or semantic content of fluorescence microscopy images. As a result, FID may provide misleading assessments, failing to capture biologically relevant differences or structural fidelity critical in cell imaging. Moreover, FID aggregates information across entire datasets, obscuring per-image and per-cell differences vital for phenotypic analysis. Therefore, we do not report FID score in our experiments.

4.2 Biological Feature via CellProfiler

To address the shortcomings of standard image metrics in biological contexts, we evaluate feature-level correspondence using CellProfiler, an open-source toolkit for cell segmentation and quantitative analysis. By extracting a high-dimensional vector of morphological, intensity, and texture features from

¹CellProfiler: <https://cellprofiler.org>

both real and synthetic images, we can directly assess whether the synthesized immunofluorescence images preserve information essential for biological interpretation.

Feature Extraction. Our analysis uses a CellProfiler pipeline² based on [3], which first segments nuclei, cytoplasm, and whole cells, and then computes a broad set of features for each segmented object. These measurements are summarized in Table 1. Bright-field channels, not generated by our model, are excluded.

Table 1: Representative feature categories extracted by CellProfiler for evaluation.

Category	Example Features
Image Quality	Multiscale blur, saturation, intensity stats, thresholding
Colocalization	Pearson r , Manders’, rank-weighted coefficients
Granularity	Spectra at multiple scales (up to 16 px)
Intensity	Mean, total, radial intensity, Zernike moments
Size & Shape	Area, perimeter, axis lengths, Zernike descriptors
Texture	GLCM-based texture (contrast, correlation, etc.)
Neighbors	Neighbor counts, contact fractions
Counts & QC	Cell/nucleus counts, QC flags

Overall, we extract 1180 features for whole cells, 1165 for cytoplasm, and 1173 for nuclei per image. These descriptors form the basis for quantitative comparison between real and synthesized data.

4.2.1 Object Matching for Single-Cell Analysis

A key challenge in our evaluation is establishing accurate correspondence between individual cells in real and synthetic images. Stochasticity in segmentation and generation process can cause variations in both cell count and spatial arrangement, making direct comparison of per-cell features difficult. Figure 6 illustrates that, even when real and generated images appear visually similar, automated segmentation can produce differing cell boundaries and counts. Therefore, an explicit matching step is necessary to align cells (or nuclei) prior to computing feature correlations.

Matching Algorithms. To establish per-object correspondences, we propose two matching strategies:

- **Centroid Matching:** Each synthetic cell is paired with its nearest real cell based on Euclidean distance between their centroids, subject to a maximum distance threshold.
- **IoU Matching:** The intersection-over-union (IoU) between all real-synthetic mask pairs is computed, and the optimal one-to-one assignment is found via the Hungarian algorithm (`scipy.optimize.linear_sum_assignment`), retaining only matches above an IoU threshold.

IoU matching, while more computationally demanding, yields more reliable pairs in dense or morphologically variable settings and serves as our primary method. Centroid matching provides a fast alternative for large-scale evaluation.

²Pipeline available at https://github.com/jump-cellpainting/2024_Chandrasekaran_NatureMethods/tree/main/pipelines/2020_11_04_CPJUMP1

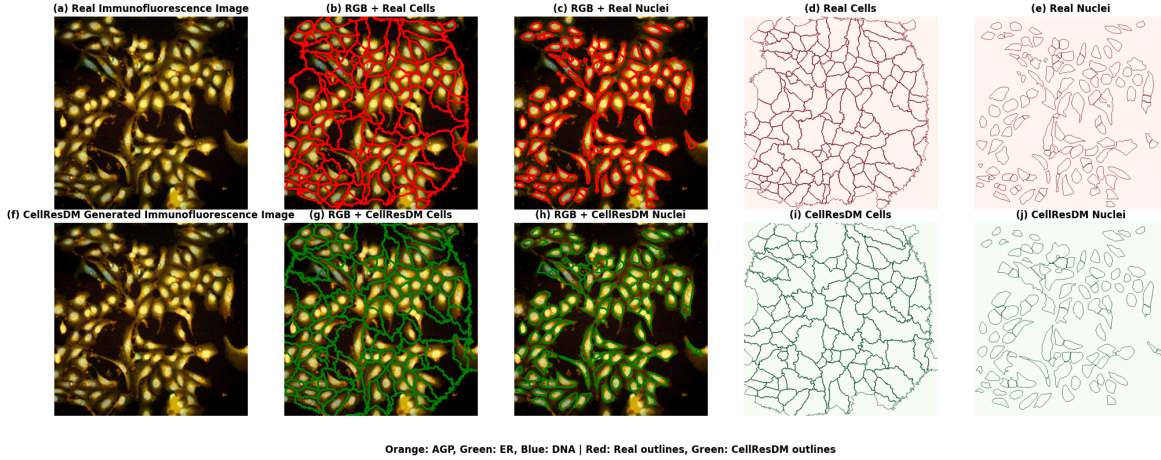


Figure 6: **CellProfiler segmentations on real and CellResDM-generated immunofluorescence images.** (a, f): Raw images; (b, g): boundaries; (c, h): nuclei; (d, i), (e, j): segmentation outlines for cells/nuclei, real (top) and synthetic (bottom). Red: real segmentations; Green: CellResDM segmentations. Despite high visual similarity, segmentation outcomes differ, motivating the need for explicit object matching.

4.2.2 Feature Correlation Computation

With matched cell pairs established between real and synthetic images, we quantify morphological fidelity by computing the Pearson correlation for each feature across all matched cell pairs. We report the mean and standard deviation of these feature-wise correlations as an overall measure of how well each synthetic method preserves single-cell morphology.

4.2.3 Unsupervised Feature Space Visualization

Beyond per-feature correlations, we qualitatively assess the preservation of global morphological structure using unsupervised dimensionality reduction. Specifically, we jointly embed the high-dimensional CellProfiler feature vectors from both real and synthesized images into a two-dimensional space via Uniform Manifold Approximation and Projection (UMAP) [15]. By visualizing the combined feature distributions, we can directly evaluate whether synthetic cells occupy similar regions of morphological feature space as real cells, and whether the diversity of cellular phenotypes is faithfully reproduced. Meaningful overlap between the real and synthetic point clouds in UMAP space indicates that the generative model successfully captures the complex, population-level variability and inter-feature relationships observed in real data. Conversely, systematic shifts, fragmentation, or mode collapse in the synthetic distribution may highlight areas where biological fidelity is lacking.

By combining standard image quality metrics with detailed, cell-level biological feature analysis, our evaluation protocol ensures that models are judged not only by their ability to reconstruct images at the pixel and perceptual level, but also by their capacity to preserve information critical for scientific discovery.

5 Dataset and Experimental Setting

5.1 Dataset

We evaluated the methods on the public **cpg0000** dataset from the Cell Painting Gallery [3], comprising over three million paired bright-field and immunofluorescence images. For robust evaluation, we curated a representative subset of eight plates, stratified by perturbation type and cell line (Table 2). Of these plates, four were used for training, two for validation, and two for testing. To assess the generalisability of our models, two additional plates under distinct CRISPR perturbations were reserved exclusively for out-of-distribution evaluation.

Imaging Modalities.

- **Bright-field image:** Each field of view is captured at three z -planes: the central focal plane (Bright-field), $5\mu\text{m}$ above (BFHigh), and $5\mu\text{m}$ below (BFLow). We use only the central Bright-field channel in our main experiments to ensure simplicity and comparability with prior work.
- **Immunofluorescence image:** Five fluorescent channels highlight distinct cellular components according to the Cell Painting protocol: AGP (actin, Golgi, plasma membrane), DNA (nucleus), ER (endoplasmic reticulum), Mito (mitochondria), and RNA (nucleoli and cytoplasmic RNA).

Plate Barcode	Perturbation	Cell Type	Num. Images	Purpose
BR00116991	compound	A549	27,648	Train
BR00116992	compound	A549	27,640	Train
BR00116995	compound	U2OS	27,648	Train
BR00117024	compound	U2OS	27,648	Train
BR00116993	compound	A549	27,352	Validation
BR00117025	compound	U2OS	27,648	Validation
BR00116994	compound	A549	27,576	Test
BR00117026	compound	U2OS	27,648	Test
BR00118041*	crispr	A549	27,560	OOD Test
BR00118045*	crispr	U2OS	27,648	OOD Test

Table 2: Summary of plates and data splits from the cpg0000 dataset. Plates marked with an * were reserved exclusively for out-of-distribution (OOD) generalization evaluation.

5.2 Experimental Setting

Data Preprocessing. Prior to training and inference, all images are uniformly resized to 512×512 pixels. Contrast is enhanced by clipping the lowest and highest 5% of pixel intensities in each image. Pixel values are normalized to $[0, 1]$ for training, while $[0, 255]$ scaling is used for visualization and final evaluation.

Model Architecture. Our core architecture is a U-Net augmented with Swin Transformer modules (UNetModelSwin) [14]. The model concatenated bright-field, y_0 , and x_t and predicts the five-channel immunofluorescence output. Key architectural choices include 160 base channels, spatial attention at resolutions $\{64, 32, 16, 8\}$, two residual blocks per stage, Swin layers with depth 2 and 192-dimensional embeddings, window size 8, and MLP ratio 4.

Noise Scheduling and Training Objective. We train our diffusion model with the non-uniform geometric noise schedule described in Section 2.3 ($\kappa = 2.0$, $p = 0.3$, $\eta_T = 0.99$, $T = 15$ steps). To stabilize optimization, we omit the explicit loss coefficient and supervise learning with a weighted combination of MSE and LPIPS [24] losses, as detailed in Section 2.2.

Baselines. We benchmark our approach against three complementary baselines:

- **U-Net** [16]: A widely adopted baseline for image-to-image prediction.
- **Pix2Pix** [10]: A conditional GAN, outperforming more advanced architectures (e.g., SPADE, SPADE-Diffusion, PLATTE) for Cell Painting synthesis as shown in [22].
- **DeepHCS** [13]: A two-stage network specifically designed for immunofluorescence synthesis, where the first stage translates bright-field to immunofluorescence, and the second stage de-noises the prediction.

These baselines together span deterministic, generative, and task-specific generative modelling approaches.

Ablation Studies. To dissect the contribution of each modelling component, we perform three ablation test:

1. *LPIPS Loss*: We train with only MSE supervision to assess the benefit of perceptual loss.
2. *Loss Coefficient*: We reintroduce the time-step-dependent loss coefficient to measure its impact on training stability and image quality.
3. *Latent Diffusion*: We encode five-channel immunofluorescence images with a pre-trained SD-VAE, splitting them into two three-channel groups, and train the diffusion process in this latent space to evaluate effects on reconstruction fidelity.

Consistency Distillation. To accelerate inference, we apply consistency distillation to our fully trained residual diffusion model. Fine-tuning is performed for one epoch with an EMA decay of 0.9999. At inference, we made the distilled model to predict with only two reverse-diffusion steps.

Training Protocol. All experiments use a fixed seed of 42, learning rate 5×10^{-5} , cosine learning-rate schedule with 5,000 warm-up steps, and are run for 10 epochs. Batch size is 2 per device for training and 4 for evaluation; no weight decay is used.

6 Experimental Results

In this section, we present a comprehensive evaluation of our proposed methods and baselines, following the evaluation described in Section 4.

6.1 quantitative Results

We first assess the pixel-wise accuracy of each model: MSE, PSNR, SSIM. Each metric is computed independently for all five immunofluorescence channels, with results aggregated across the entire test set.

The detailed results are summarized in Table 3 and visualized as channel-wise distributions in Figure 7.

Baseline Performance. Among all baselines, **DeepHSC** achieves the highest overall quantitative scores, outperforming both the U-Net and Pix2Pix models as well as the original CellResDM in terms of lower MSE and higher SSIM/PSNR. While it is somewhat unexpected for DeepHSC to marginally outperform CellResDM in some channel, the performance gap remains modest.

Effect of Consistency Distillation. Notably, the **Consistency Distilled CellResDM** inferred with 2-step achieves a substantial performance boost over all other models, recording the lowest average MSE and the highest SSIM and PSNR across most channels. This demonstrates that consistency distillation not only accelerates inference but also enhances reconstruction fidelity, consistent with our hypothesis.

Channel-wise Trends. Across all models, the DNA channel consistently yields the best reconstruction metrics—highest SSIM and PSNR, and lowest MSE. This advantage arises because, DNA staining results in compact, spatially localized “blobs” that are distinctly separated for each cell, making DNA morphologically simple and straightforward to predict. In contrast, the Mito, AGP, RNA, and ER channels capture a range of cytoplasmic and organelle features, as well as some nuclear details. These compartments are inherently more complex: they exhibit highly textured patterns, diffuse and sometimes overlapping boundaries, and significant heterogeneity both within and across cells. As a result, these channels present a much greater challenge for accurate synthesis and morphological fidelity in all models.

Model	MSE ₂₅₅ ↓	SSIM ₂₅₅ ↑	PSNR ₂₅₅ ↑
DeepHSC	617.9 ± 473.6	0.794 ± 0.103	20.85 ± 2.14
U-Net	1140.5 ± 1908.3	0.622 ± 0.101	18.99 ± 2.69
Pix2Pix	1626.1 ± 829.5	0.433 ± 0.087	16.44 ± 1.83
CellResDM (15 steps)	734.6 ± 543.3	0.728 ± 0.119	20.24 ± 2.49
CellResDM (2 steps, Consistency Distilled)	568.7 ± 1136.1	0.768 ± 0.126	21.98 ± 2.58

Table 3: Mean ± standard deviation of quantitative metrics across the test set for each model. Lower MSE and higher SSIM/PSNR indicate improved reconstruction quality.

6.2 Ablation Results

To clarify the necessity of each component in our framework, we did ablation study on three aspects of CellResDM: (i) the LPIPS loss, (ii) omission of the loss coefficient, and (iii) latent diffusion via SD-VAE. Figure 8 and Table 4 summarize the results. In every case, these ablations significantly degrade performance relative to the baseline, confirming the importance of each design choice.

Model	MSE ₂₅₅ ↓	SSIM ₂₅₅ ↑	PSNR ₂₅₅ ↑
CellResDM (15 step)	734.6 ± 543.3	0.728 ± 0.119	20.24 ± 2.49
CellResDM (no LPIPS)	1573.8 ± 803.0	0.549 ± 0.100	16.65 ± 2.06
CellResDM (with loss coefficient)	878.0 ± 509.0	0.689 ± 0.108	19.16 ± 1.91
CellResDM (SD-VAE latent diffusion)	1451.3 ± 521.5	0.475 ± 0.089	16.74 ± 1.39

Table 4: Mean ± standard deviation of metrics for each ablation variant (lower MSE, higher SSIM/PSNR are better).

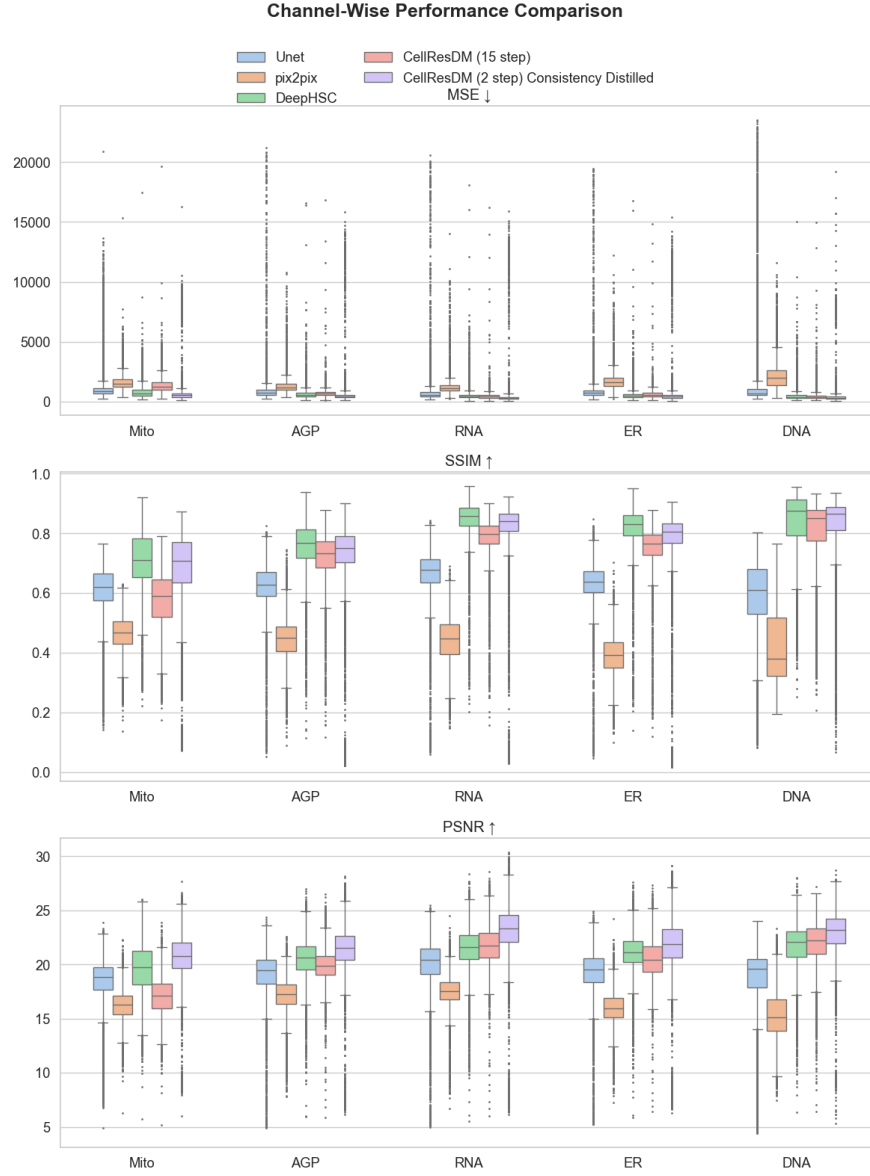


Figure 7: **Channel-wise comparison of reconstruction metrics** (MSE, SSIM, PSNR) for all evaluated models. Boxplots display the distribution for each channel (Mito, AGP, RNA, ER, DNA) over the test set. CellResDM and Consistency Distilled CellResDM consistently outperform other baselines, with the DNA channel being the most reliably reconstructed.

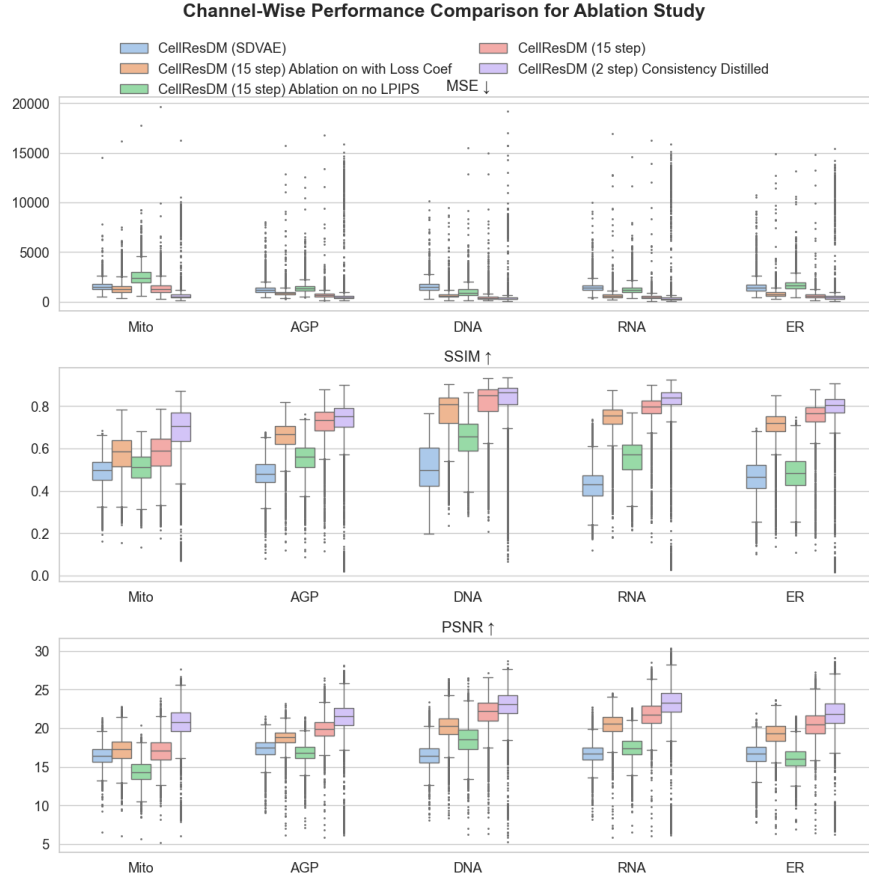


Figure 8: Quantitative ablation results for five imaging channels (Mito, AGP, DNA, RNA, ER). Boxplots show MSE (lower is better), SSIM, and PSNR (both higher are better). All ablated variants underperform both baseline and consistency-distilled CellResDM.

LPIPS Loss. Omitting LPIPS from the loss worsens all metrics because MSE alone only penalizes pixel-wise errors. LPIPS, though trained on natural images, still helps by supervising deep features—guiding the model to preserve texture and structure beyond what pixel-wise loss can capture.

Loss Coefficient. When reintroduced, the imbalanced loss coefficient produces unstable training, as evidenced by the loss curve in Figure 9, and consistently degrades reconstruction quality.

Latent Diffusion. We further evaluated CellResDM in a latent diffusion setting using a pretrained SD-VAE encoder³. SD-VAE can reconstruct cell images with very low MSE and no fine-tuning (Figure 10), confirming its suitability for the domain [2]. However, applying residual diffusion in latent space resulted in much poorer performance. This aligns with prior observations [22], likely because the linear residual shift in pixel space becomes a complex, non-linear mapping in the SD-VAE latent space, substantially complicating optimization and reducing fidelity.

³SD-VAE model weight: <https://huggingface.co/stabilityai/sd-vae-ft-mse>

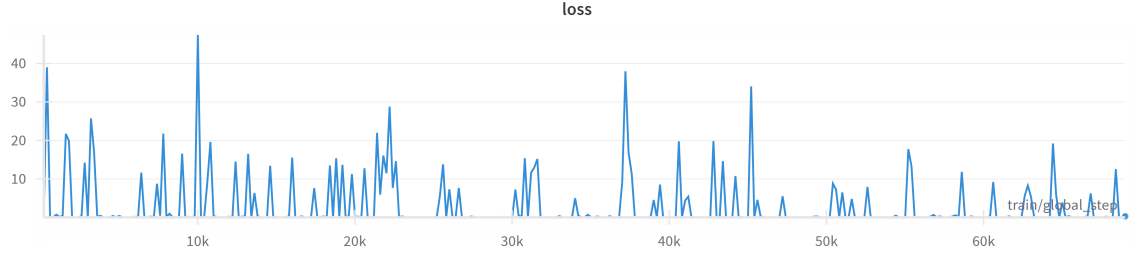


Figure 9: Loss curve for CellResDM with reintroduced loss coefficient, showing instability and poor convergence.

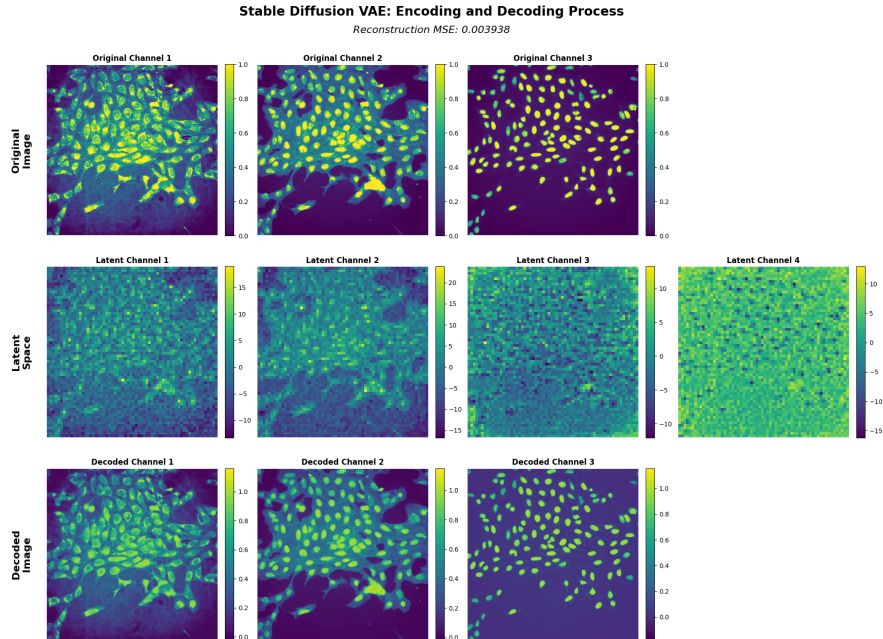


Figure 10: SD-VAE reconstructions (bottom row) of cell images (top row) closely match the originals, but latent diffusion fails to achieve similar fidelity.

6.3 In- vs. Out-of-Distribution Performance

Table 5 summarizes evaluation results on in-distribution (ID) and out-of-distribution (OOD) data. Δ indicates performance change (OOD – ID); lower MSE and higher SSIM/PSNR reflect better results. **DeepHSC** suffers the greatest performance drop on OOD data, highlighting poor generalization. In contrast, **Consistency-Distilled CellResDM (2-step)** remains the most robust and efficient, with OOD MSE even improving.

6.4 Qualitative Results

We present a qualitative comparison of real and synthetic images for all models in Figures 11 and 12. As shown in Figure 11, Unet predictions appear blurry with weak cell boundaries, while Pix2Pix is high in error. DeepHSC, CellResDM and Consistency Distilled CellResDM stand out for their close visual match to real images, as evidenced by minimal difference in the error maps.

Table 5: In-distribution (ID) and out-of-distribution (OOD) results for all models. Δ shows the change from ID to OOD (positive MSE and negative SSIM/PSNR values indicate worse OOD performance). Best ID scores are **bolded**.

Model	MSE			SSIM			PSNR		
	ID	OOD	Δ	ID	OOD	Δ	ID	OOD	Δ
CellResDM-15	648	821	+173	0.746	0.709	-0.038	20.69	19.80	-0.89
CellResDM-2 (+CD)	609	521	-88	0.780	0.752	-0.028	22.43	21.43	-1.00
DeepHSC	556	679	+123	0.827	0.760	-0.067	21.34	20.36	-0.98
U-Net	1065	1216	+151	0.614	0.630	+0.016	19.24	18.74	-0.51
Pix2Pix	1595	1658	+63	0.414	0.452	+0.038	16.50	16.37	-0.13

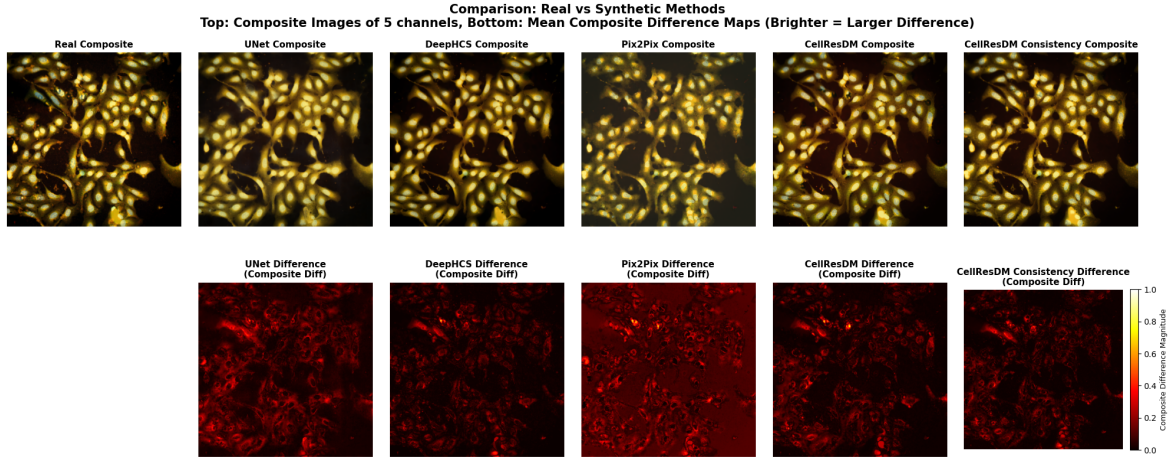


Figure 11: Top: Composite images of all five channels for real and synthetic data. Bottom: Mean difference maps (brighter = greater deviation from real). CellResDM and its consistency-distilled variant show the smallest errors, indicating high fidelity.

Figure 12 provides zoomed-in views to highlight fine structural details. Unet’s outputs are over-smoothed, losing key nuclear features and cell boundaries. Pix2Pix frequently hallucinates or erases cellular structures, leading to inconsistent results. DeepHCS often omits portions of cells and still produces overly smoothed morphologies. In contrast, both CellResDM and its consistency-distilled variant accurately reconstruct both global morphology and fine cellular detail, consistently matching the real images even in the most challenging regions.

These qualitative results reinforce the quantitative findings: CellResDM and its consistency-distilled variant offer the highest visual fidelity and structural accuracy among all evaluated methods.

6.5 Morphological Feature Fidelity

While high-level image similarity metrics provide a useful baseline, they do not guarantee that synthetic images retain biologically meaningful structure at the single-cell level. To directly assess the biological realism of each synthesis model, we quantified the preservation of *single-cell morphological features* extracted using CellProfiler. Correlations were computed between matched real-synthetic object pairs (Sec. 4.2.1), summarizing how faithfully each method preserves cell-level phenotypes.

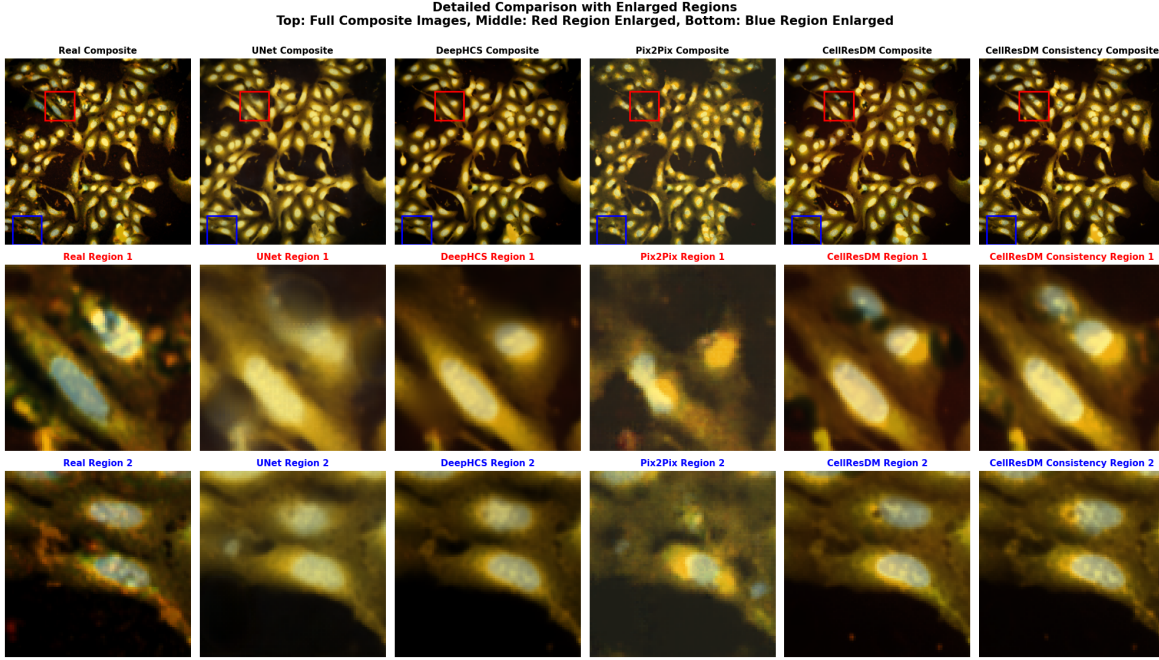


Figure 12: Enlarged qualitative comparison. Top: Full composites; middle/bottom: Zoom-ins of red- and blue-boxed regions. CellResDM and its consistency-distilled variant best preserve fine cellular structures; baselines often lose detail or introduce artifacts.

Global performance. Figure 13 visualizes feature correlations by compartment; Table 6 summarizes the results for each model. The consistency-distilled **CellResDM** achieves the highest overall correlation ($r = 0.565$), narrowly surpassing **DeepHCS** ($r = 0.555$) and the original 15-step **CellResDM** baseline. **UNet** trails with lower correlations, and **Pix2Pix** performs worst ($r = 0.314$), consistent with its qualitative artifacts (Fig. 12).

Table 6: Mean Pearson r by compartment (numeric summary of Fig. 13). Higher is better.

Model	Cell	Cytoplasm	Nuclei	Overall
UNet	0.434	0.401	0.413	0.416
DeepHCS	0.572	0.542	0.552	0.555
Pix2Pix	0.327	0.280	0.335	0.314
CellResDM	0.527	0.494	0.574	0.532
CellResDM_Distill	0.558	0.530	0.607	0.565

Feature-specific trends.

- **Relational (Parent) features:** All models achieve near-perfect correlation ($r > 0.95$). This indicates that, despite intensity or texture errors, all nuclei remain correctly nested within their corresponding cell. Thus, no systematic or readily observable biological errors occur at the object-assignment level across any method.
- **Count-based (Number, Neighbors) features:** Correlations range from $r \approx 0.4$ to 0.7 , scaling with overall model fidelity. Pix2Pix consistently underestimates cell and nucleus counts due to

missed or merged objects, while UNet’s over-smoothing impairs neighbour relationship detection. Diffusion-based models achieve the highest correlations, though there is still room for improvement.

- **Spatial statistics (Location, Correlation):** Features capturing geometry or long-range spatial dependencies remain difficult, with no model exceeding $r = 0.40$. Pix2Pix again performs worst ($r < 0.25$), and even the best models struggle to reproduce radial or texture patterns, reflecting limitations of current loss functions for enforcing higher-order structure.



Figure 13: Correlation matrix for morphological feature groups across cell, cytoplasm, and nuclei compartments.

6.6 Channel-wise Texture Feature Correlations

We further evaluated each model on more fine-grained features—*Granularity*, *Intensity*, *Radial Distribution*, and *Texture*—across all five Cell Painting channels (Mito, AGP, RNA, ER, DNA) and three cellular compartments (cell, cytoplasm, nucleus). Each feature family comprises multiple sub-features, and results were aggregated accordingly. Figure 14 shows the correlation coefficients (r) for every channel-compartment-feature combination.

Intensity features: First-order intensity statistics are most robustly captured, with median correlations of $r \approx 0.7$ for CellResDM, DeepHSC, and their distilled variants, and $r \approx 0.5$ – 0.6 for UNet. Pix2Pix consistently performs worst, with particularly low accuracy on the DNA channel. This poor DNA channel performance may be due to the typically higher intensity values in DNA regions, which most models fail to accurately reproduce.

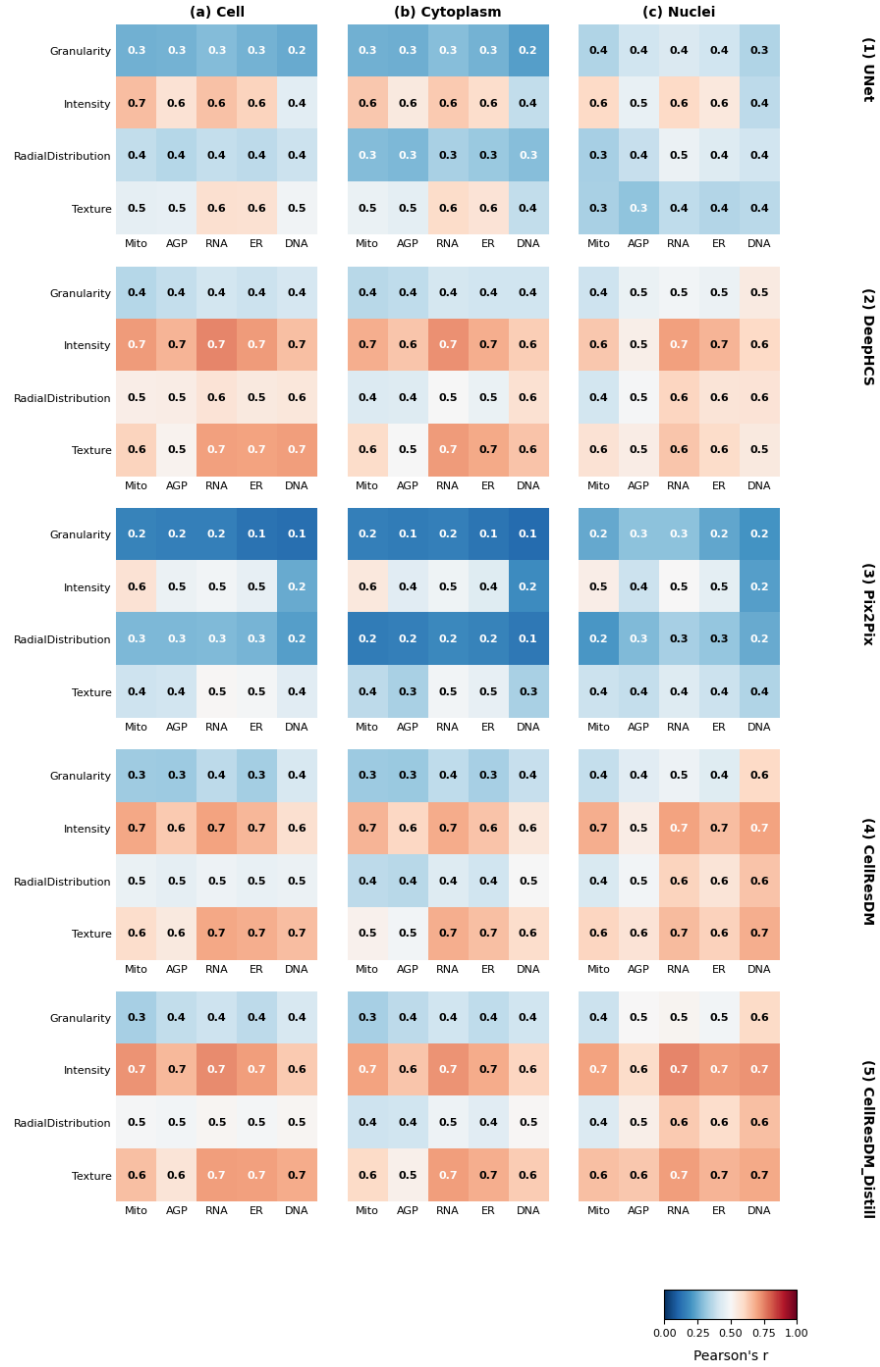


Figure 14: Correlation matrix for morphological feature groups across cell, cytoplasm, and nuclei compartments.

Granularity features: Granularity, which quantifies the presence of small, repetitive structures, shows improved correlation with more advanced architectures. While Pix2Pix achieves $r \approx 0.2$, this rises to 0.3 in UNet, and 0.4 or above for DeepHCS and diffusion-based models—especially in mitochondrial and RNA channels. However, cytoplasmic granularity remains more challenging, with generally lower scores, reflecting difficulties in capturing subtle, low-contrast textures.

Radial distribution: All models struggle with radial distribution features, which encode concentric intensity gradients and spatial organization. Even the best models rarely exceed $r \approx 0.5$ in the cell and nucleus, and fall below 0.4 in cytoplasm. Pix2Pix again performs the worst, seldom surpassing $r = 0.3$. These results highlight the limitations of current architectures and loss functions in modelling long-range spatial dependencies.

Texture features fall between the above extremes. CellResDM, DeepHCS, and their distilled versions achieve correlations in the 0.5–0.7 range, with UNet somewhat lower and Pix2Pix lowest overall.

Across all feature families and imaging channels, the same ranking of model performance consistently emerges: Pix2Pix performs worst, followed by UNet, then CellResDM, with DeepHCS achieving slightly better results, and the consistency-distilled CellResDM variant achieving the highest correlations overall. The distilled two-step CellResDM generally matches or outperforms CellResDM, confirming that efficiency gains do not come at the cost of biological fidelity.

Taken together, these findings demonstrate that while modern diffusion and transformer-based methods reproduce first-order and medium-scale texture statistics well, further advances are needed to fully capture higher-order spatial and organizational features in cellular images.

6.7 UMAP Visualization of Morphological Feature Distributions

While feature correlations quantify cell-level fidelity, they do not capture the global structure of the feature distributions. To visualize this, we jointly embed CellProfiler feature vectors from real and synthetic images using UMAP [15].

Figure 15 shows UMAP projections for whole-cell, cytoplasmic, and nuclear features. Each point represents a cell or nucleus coloured by synthesis method. Strong overlap with the real distribution indicates high morphological realism.

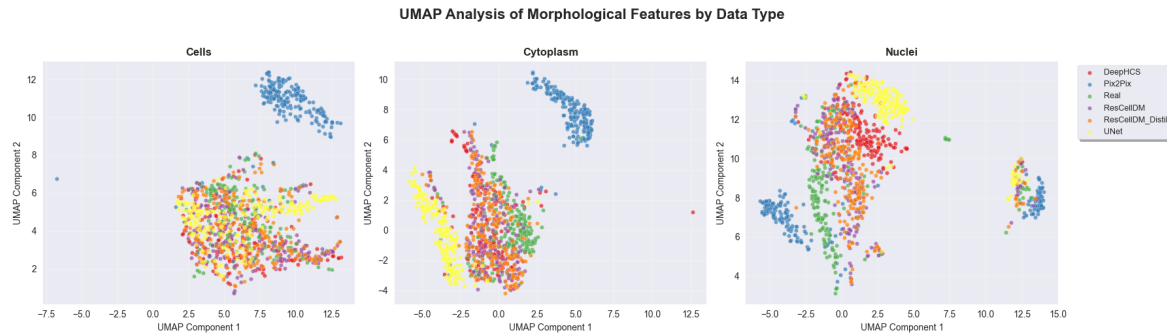


Figure 15: UMAP projections of CellProfiler features for whole cells, cytoplasm, and nuclei. Each point is a cell or nucleus from real or synthetic images, coloured by method. Overlap indicates preservation of population-level morphology.

In all three compartments, **CellResDM** and its consistency-distilled variant show near-complete overlap with the real data, faithfully capturing the full diversity of cellular morphologies. **DeepHCS** also aligns closely but cluster slightly above the main real-cell manifold in the nucleus panel, indicating minor deviations in nuclear detail. **UNet** exhibits noticeable shifts and broader dispersion, reflecting its tendency to over-smooth boundaries and blur textures. **Pix2Pix** forms distinct, peripheral clusters separated from the real distribution, consistent with its frequent loss or misplacement of cells

and structures. Overall, these patterns reinforce that only CellResDM and Consistency Distilled CellResDM generate synthetic populations that truly mirror the morphological complexity of real cells, while others distort population structure and reduce biological fidelity.

7 Discussion and Conclusion

This work reproduced CellResDM, a residual-shifted diffusion framework for brightfield to immunofluorescence image synthesis, and demonstrates its superior performance over established baselines. Our comprehensive evaluation shows that CellResDM, especially when combined with consistency distillation, consistently outperforms DeepHCS, Pix2Pix, and U-Net—both in standard image metrics and biological fidelity. Pix2Pix often hallucinates structures and generates incomplete or fragmented cellular features, while U-Net tends to produce overly smoothed, blurred images that lack fine morphological details. DeepHCS, despite strong in-distribution performance, exhibits poor generalization and suffers significant degradation on out-of-distribution data. In contrast, CellResDM achieves lower MSE and higher SSIM/PSNR, and excels at preserving biologically meaningful morphological features, as evidenced by high correlation with ground-truth single-cell metrics and clear clustering in UMAP visualizations. Overall, CellResDM provides the best and most balanced performance across both technical and biological criteria, making it a robust and reliable solution for label-free cell profiling.

Consistency distillation proves especially impactful: reducing inference from 15 to just 2 steps yields significant acceleration without sacrificing fidelity—indeed, we even observe slight improvements in both global image quality and morphological preservation. Our ablation studies further reveal that each component of the CellResDM is necessary; perceptual loss, balanced noise scheduling, and direct pixel-space modelling each contribute to stable optimization and faithful reconstruction.

Nonetheless, certain limitations persist. Capturing fine-grained texture and spatial distribution features, such as radial gradients or higher-order texture cell features, remains challenging, indicating room for improvement in future architectures or loss designs. Furthermore, although CellResDM generalizes well to held-out and out-of-distribution conditions within the current dataset, evaluation on rarer cell types or diverse imaging protocols will be essential to fully validate robustness.

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A Mathematical Derivations for CellResDM

In this appendix, we provide detailed mathematical derivations and proofs that underpin the main methods described in Section 2. These include the reparametrization of the forward process by induction, the computation of the true posterior for the reverse process, and the derivation of the per-step KL loss coefficient for training. Readers less interested in technical derivations may safely skip to the main results, but these details may be useful for those seeking to reproduce or extend our work.

A.1 Forward Process Reparametrization by Induction

As described in Section 2, the forward noising process can be parametrized to depend directly on the initial images x_0 and y_0 . We now provide the induction proof supporting Equation (2).

Base Case ($t = 1$):

$$\begin{aligned} q(x_1|x_0, y_0) &= \mathcal{N}(x_1; x_0 + \alpha_1 e_0, \kappa^2 \alpha_1 I) \\ &= \mathcal{N}(x_1; x_0 + \eta_1 e_0, \kappa^2 \eta_1 I) \quad (\text{since } \alpha_1 = \eta_1) \end{aligned}$$

which matches Eq. (2) for $t = 1$.

Inductive Step: Assume $q(x_t|x_0, y_0) = \mathcal{N}(x_t; x_0 + \eta_t e_0, \kappa^2 \eta_t I)$ holds for some t . Then,

$$\begin{aligned} x_{t+1} &= x_t + \alpha_{t+1} e_0 + \kappa \sqrt{\alpha_{t+1} \epsilon_{t+1}} \\ &= (x_0 + \eta_t e_0 + \kappa \sqrt{\eta_t \epsilon_t}) + \alpha_{t+1} e_0 + \kappa \sqrt{\alpha_{t+1} \epsilon_{t+1}} \\ &= x_0 + (\eta_t + \alpha_{t+1}) e_0 + \kappa \sqrt{\eta_t \epsilon_t} + \kappa \sqrt{\alpha_{t+1} \epsilon_{t+1}} \\ &= x_0 + \eta_{t+1} e_0 + \kappa \sqrt{\eta_t \epsilon_t} + \kappa \sqrt{\alpha_{t+1} \epsilon_{t+1}} \end{aligned}$$

The sum of two independent Gaussians is Gaussian with variance $\kappa^2(\eta_t + \alpha_{t+1})I = \kappa^2 \eta_{t+1} I$, thus:

$$x_{t+1} \sim \mathcal{N}(x_0 + \eta_{t+1} e_0, \kappa^2 \eta_{t+1} I)$$

completing the induction.

A.2 True Posterior for the Reverse Process

The main text (Section 2) gives the form of the true reverse posterior, which is essential for deriving the training objective. Here, we show its derivation by completing the square for two Gaussian distributions.

Recall:

$$\begin{aligned} q(x_t|x_{t-1}, y_0) &= \mathcal{N}(x_t; x_{t-1} + \alpha_t e_0, \kappa^2 \alpha_t I) \\ q(x_{t-1}|x_0, y_0) &= \mathcal{N}(x_{t-1}; x_0 + \eta_{t-1} e_0, \kappa^2 \eta_{t-1} I) \end{aligned}$$

By Bayes' theorem and the Markov property, the true reverse posterior is:

$$\begin{aligned} q(x_{t-1}|x_t, x_0, y_0) &= \frac{q(x_t|x_{t-1}, y_0) q(x_{t-1}|x_0, y_0)}{q(x_t|x_0, y_0)} \\ &\propto q(x_t|x_{t-1}, y_0) q(x_{t-1}|x_0, y_0) \end{aligned}$$

Multiplying these two Gaussians (with respect to x_{t-1}) and completing the square, we obtain:

$$q(x_{t-1}|x_t, x_0, y_0) = \mathcal{N}(x_{t-1}; \mu_t, \Sigma_t)$$

with

$$\begin{aligned} \mu_t &= \frac{\eta_{t-1}}{\eta_t} x_t + \frac{\alpha_t}{\eta_t} x_0 \\ \Sigma_t &= \kappa^2 \frac{\eta_{t-1}}{\eta_t} \alpha_t I \end{aligned}$$

where $\eta_t = \eta_{t-1} + \alpha_t$.

A.3 KL Loss Coefficient Derivation

The per-step training loss for CellResDM is based on the KL divergence between the true posterior and the model's reverse step (see Section 2). Here, we expand the derivation.

Let:

$$\begin{aligned} q(x_{t-1}|x_t, x_0, y_0) &= \mathcal{N}(x_{t-1}; \mu_q, \Sigma_t) \\ p_\theta(x_{t-1}|x_t, y_0) &= \mathcal{N}(x_{t-1}; \mu_\theta, \Sigma_t) \end{aligned}$$

where

$$\begin{aligned} \mu_q &= \frac{\eta_{t-1}}{\eta_t} x_t + \frac{\alpha_t}{\eta_t} x_0 \\ \mu_\theta &= \frac{\eta_{t-1}}{\eta_t} x_t + \frac{\alpha_t}{\eta_t} f_\theta(x_t, y_0, t) \end{aligned}$$

The KL divergence between two Gaussians with the same covariance is:

$$D_{\text{KL}}(q(\mu_q, \Sigma_t) \| p_\theta(\mu_\theta, \Sigma_t)) = \frac{1}{2} (\mu_q - \mu_\theta)^\top \Sigma_t^{-1} (\mu_q - \mu_\theta)$$

Computing the mean difference:

$$\mu_q - \mu_\theta = \frac{\alpha_t}{\eta_t} (x_0 - f_\theta(x_t, y_0, t))$$

and

$$\Sigma_t = \kappa^2 \frac{\eta_{t-1}}{\eta_t} \alpha_t I \implies \Sigma_t^{-1} = \frac{1}{\kappa^2} \frac{\eta_t}{\eta_{t-1}} \frac{1}{\alpha_t} I$$

Plugging these in:

$$\begin{aligned} D_{\text{KL}} &= \frac{1}{2} \left(\frac{\alpha_t}{\eta_t} (x_0 - f_\theta) \right)^\top \frac{1}{\kappa^2} \frac{\eta_t}{\eta_{t-1}} \frac{1}{\alpha_t} \left(\frac{\alpha_t}{\eta_t} (x_0 - f_\theta) \right) \\ &= \frac{1}{2\kappa^2} \frac{\alpha_t}{\eta_{t-1}\eta_t} \|x_0 - f_\theta(x_t, y_0, t)\|_2^2 \end{aligned}$$

Therefore, the full per-timestep KL term becomes:

$$D_{\text{KL}} = \frac{\alpha_t}{2\kappa^2\eta_{t-1}\eta_t} \|x_0 - f_\theta(x_t, y_0, t)\|_2^2$$

As discussed in Section 2, this coefficient is omitted in practice due to instability for small η_{t-1} .

B Differences from the Original CellResDM

During the reproduction of CellResDM, several differences arose due to design choices and computational constraints.

Correlation Calculation: In our approach, we implemented a cell matching step prior to correlation analysis. The original paper [21] computes the mean of each feature group and then calculates the correlation. While mean is a useful statistical method, we believe it can obscure important details. By matching individual cells, we are able to perform per-cell, per-feature correlation analysis, allowing for a more granular assessment.

Training Dataset: The original authors excluded images at the edges of the plates, as these contained no useful information. However, we were unable to reliably identify edge images, and not all edge images in our dataset were uninformative. Consequently, we did not exclude any such images. This difference may partly explain why our results are slightly worse than those reported in the original CellResDM paper.

MoA Performance Metrics (Table 3 in [21]): We were unable to evaluate predictive performance for mechanism-of-action (MoA) classification. This limitation arose because CellProfiler could not be run on the cloud platform (CSD3), and local processing constrained the number of images we could analyze. Consequently, we did not have sufficient CellProfiler-extracted features to train an SVM, and were therefore unable to reproduce this part of the original evaluation.

Segmentation Task: Due to strict storage limitations on CSD3 (1TB, which was already largely occupied by the brightfield and immunofluorescence images), we were unable to generate and store the additional segmentation masks required for both cell-level and nuclei-level analyses. This prevented us from reproducing the segmentation results shown in [22].

C AI Usage

For this report, I used ChatGPT to refine grammar and improve clarity. Additionally, Claude assisted with coding tasks, including generating more visually appealing plots, partially writing the training script and README, and providing code auto-completion.